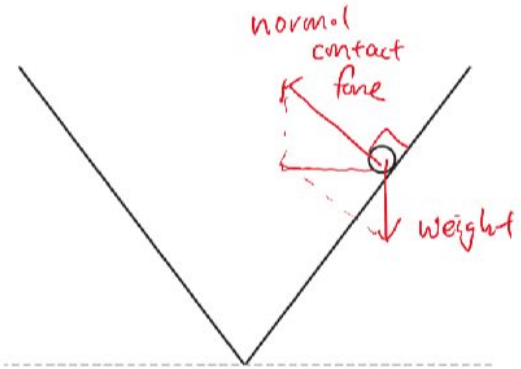
2(a)	 correct arrow direction – vertically downwards arrow for weight and arrow perpendicular to the cone, inwards and upwards for normal contact correct naming for both forces – weight and normal contact force/ normal force	 [1] [1]
2(b)	vertical component of the normal contact force = weight, thus F_{net} = 0 in the vertical direction horizontal component of the normal contact force is resultant force towards centre (of circle)	 [1] [1]
2(c)	N cos52° = W = mg ----- (1) N sin52° = mv^2/r ----- (2) (2)/(1): tan 52° = v^2/gr v = gr tan 52° = 9.81 (0.15)(tan 52°) = 1.372 = 1.4 m s^{-1} (shown)	[1] [1] [1]
2(d)	v = rω or a = $r\omega^2$ ω = 1.4 / 0.15 or √(9.81 x tan 52° / 0.15) = 9.3 rad s^{-1} or 9.149 = 9.1 rad s^{-1}	 [1]
2(e)	Since tan 52° = v^2/gr, v^2 is proportional to r so if speed increase, radius must also increase or (same mass) resultant/centripetal force/acceleration (N sin52° and N = W/ cos52°) = constant, since F = mv^2/r or a = v^2/r, v^2 is proportional to r so if speed increase, radius must also increase	[1] [1] [1] [1]